

COURSE 2B - DA-to-DE Bridge

B06 Bridge Capstone & Course 3 Entry

Prove the bridge, then hand off cleanly to engineering

Fast-track purpose

B06 is the evidence boundary between competent analyst work and structured Data Engineering training. It integrates B01-B05 in one genuine capstone, then uses a protected Final Gate plus competency floors to decide whether C3 may begin. It does not steal C3-00 implementation topics into the Bridge.

Locked learning loop

Why -> picture -> new words -> source/visual -> analyst-to-engineer transition -> follow once -> observe -> explain -> mistakes -> recover -> fresh task -> validate -> teach back -> protected review -> changed retry -> evidence/mastery + next route.

INTERNAL QA RC1 - not Course 2B FINAL

Contents & Study Route

Lesson	Title	Pages
B06-L01	Integrated DA-to-DE bridge capstone	3-4
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Exact open order

B06-L01 learner book -> Capstone Evidence Workbook -> SQL workspace -> Python notebook -> save complete attempt -> protected capstone review -> changed transfer -> B06-L02 -> Final Gate Rules -> Gate A -> save attempt -> matching protected Gate review -> readiness scorecard -> C3 handoff.

Protected rule

Gate B/C are sealed. A failure never auto-opens the next Gate. Narrow weakness routes to targeted repair. A fresh full Gate opens only when the review explicitly assigns reassessment.

C3 boundary

Linux/WSL2, shell, Git, venv/dependency pinning, Python packages/project structure, configuration/secrets, structured logging/exceptions, tests, CLI design, Docker and idempotence/runbooks are taught in C3-00 and are not B06 pass requirements.

B06-L01 - Integrated DA-to-DE bridge capstone

1. Why it matters

Course 3 should not begin because you finished pages. It should begin because you can connect SQL, Python, modeling, warehousing, data quality and platform reasoning into one auditable flow and recover when that flow fails.

2. What you are learning

Build and defend the Home Services Network capstone as one evidence chain: source contracts -> SQL -> Python -> model/history -> warehouse/quality -> platform/monitoring -> reconciliation/recovery -> professional handoff.

3. Prerequisite

Course 1 plus B01-B05 evidence. You are allowed to use the B01-B05 learner books during this guided capstone. The protected Final Gate comes later and is closed to solution help.

4. Picture it in your head

Picture a relay race. SQL, Python, modeling and cloud reasoning are not separate trophies. Each runner must hand trustworthy evidence to the next. If one handoff loses rows, changes grain or ignores a failed control, the whole pipeline is untrusted.

5. New words before use

evidence chain	saved proof connecting source assumptions, transformations, controls and final output
trusted publish	output released only after required controls pass
independent reconciliation	a separate check that can detect an error in the main transformation
safe rerun	running unchanged inputs again without duplicating or corrupting trusted business rows
failure recovery	detect -> contain -> diagnose -> repair -> rerun/backfill -> validate -> document
handoff	clear artifacts and explanation another engineer can continue safely

6. Watch / visual source

No new external video is assigned here. This is deliberate: B06-L01 tests transfer of B01-B05, not another tutorial dependency. Use the course-built architecture/evidence map and your verified B01-B05 sources only when a specific prior skill needs review.

ANALYST -> ENGINEER TRANSITION

What you already know as an analyst: You already know how to produce an analysis using SQL/Python/Power BI and explain a business result.

What changes in engineering: Engineering requires the entire path to be reproducible, grain-safe, validated, observable enough to diagnose, and recoverable when something breaks.

Why engineers care: A correct dashboard at the end cannot prove that rows were not silently multiplied, history was not rewritten, a critical defect was not ignored, or the next rerun will not duplicate trusted data.

Small worked example: Home Services has 72 order rows, 144 item rows and 271 event rows. An engineer keeps those grains explicit, aggregates items before joining order-level fees, detects the one orphan event, blocks malformed numeric input, proves historical technician logic, and records safe rerun evidence before calling the output trusted.

Real engineering version: C3 will implement deeper engineering environments, ingestion/state, dbt, orchestration and scale. B06 proves you can enter that course with the prerequisite reasoning already stable.

7. Follow with me once

Open B06_CAPSTONE_EVIDENCE_WORKBOOK_RC1.xlsx -> START_HERE. Then open Source_Contracts. Follow the exact order shown there. Run B06_PRACTICE_POSTGRES_SQL_SETUP_RC1.sql when PostgreSQL is available, complete B06_PRACTICE_SQL_WORKSPACE_RC1.sql, then complete B06_PRACTICE_PIPELINE_NOTEBOOK_RC1.ipynb. Save every validation and failure result; do not overwrite failed attempts.

8. What you should see

Your evidence should reconcile the published order-level result to lower-grain items, show source lineage, expose rather than hide quality failures, distinguish current vs historical technician attributes, define platform/access/refresh/monitoring choices, and prove an unchanged rerun is safe.

9. Why it worked

The capstone starts from contracts and controls, not from a dashboard. That makes every later step testable: source grain -> transformation grain -> key/cardinality proof -> quality action -> publish decision -> rerun/recovery evidence.

10. Common beginner mistakes

Joining item/event/skill rows directly to order-level fees; using DISTINCT to hide fan-out; coercing BAD_AMOUNT to zero and publishing anyway; using current technician region for historical facts without a decision; treating a successful run as proof; writing a cloud diagram with no access/freshness/failure owner.

11. If stuck

Use the evidence workbook one sheet at a time. Ask: What is one row? What must be unique? What can multiply? What independent check could catch me? What blocks publish? What is saved if this fails? Who owns the next action?

12. Small version alone

After the guided Home Services attempt is complete, change the serving grain to technician-month and rebuild only the necessary SQL/model/reconciliation evidence without copying your first wording. Preserve one trusted order-value control.

13. Validate

Capstone is not complete until the workbook has evidence across all eight readiness areas, required controls reconcile, one critical failure is demonstrated and contained, an unchanged rerun is proven safe, and the handoff states one limitation plus owner/next action.

14. Teach it back

Explain the complete capstone in 90 seconds without naming tools first. Start with source grains and business contracts, then describe transformations, quality gates, history, recovery and handoff.

15. Protected review + changed retry

Save the complete first capstone attempt before opening B06_CAPSTONE_PRACTICE_REVIEW_PROTECTED_RC1.pdf. Compare only the weak area, close the review, and perform the assigned changed transfer. Append repair evidence; never erase the first attempt.

16. Evidence / mastery + next route

Required: completed evidence workbook, SQL file, notebook, validation/run evidence, failure/recovery proof, changed transfer and handoff defense. PASS -> B06-L02. Weak area -> targeted B01-B05 or B06 repair only; do not restart Course 2B.

B06-L02 - Course 3 entry readiness and handoff

1. Why it matters

A completion percentage can hide a critical gap. Course 3 assumes you can reason independently about data correctness and handoff, but it does not assume you already know Linux, Git workflows, virtual environments, Docker or other C3-00 implementation topics.

2. What you are learning

Take the protected Final Gate, score evidence against the eight readiness floors, route any weakness to the exact repair, and produce an explicit C3 handoff with no hidden prerequisites.

3. Prerequisite

B06-L01 capstone evidence complete. Gate A is the first protected assessment. Review/solution material stays closed until the attempt is saved.

4. Picture it in your head

Picture an airport gate with eight document checks. You do not board because 95% of the checklist is green if the passport check fails. But you also are not required to already know how to fly the airplane. B06 checks entry readiness; C3 teaches the next engineering environment and systems depth.

5. New words before use

readiness floor	minimum evidence level required in one competency; not an average
critical failure	a defect that blocks a pass regardless of average performance
protected attempt	independent assessment completed before opening its review
full reassessment	fresh Gate B/C only when explicitly assigned after broad/critical instability
entry ready	Final Gate PASS + every floor met + no unresolved critical blocker
hidden prerequisite	skill demanded for entry even though it was never explicitly taught beforehand

6. Watch / visual source

No external video is assigned. Open B06_C3_ENTRY_READINESS_SCORECARD_RC1.xlsx -> C3_00_Preview. It lists the 12 incoming C3-00 topics and explicitly marks each as taught in C3, not a Bridge pass requirement.

ANALYST -> ENGINEER TRANSITION

What you already know as an analyst: You already know how to judge whether an analysis is correct enough to present and how to revise weak work.

What changes in engineering: Engineering progression uses competency floors, protected independent evidence, critical-failure blockers and explicit remediation routes rather than course-completion percentages.

Why engineers care: Without floors, one strong area can mathematically hide a dangerous weakness. Without an explicit C3 boundary, learners waste time pre-learning Linux/Git/Docker or arrive believing unexplained prerequisites were missed.

Small worked example: Scores SQL=3, Python=3, Modeling=3, Warehousing=3, Quality=3, Independent=3, Cloud=2, Ownership=2 plus Final Gate PASS -> C3 ENTRY READY. If Python=2, overall status stays NOT READY even if every other score is 4.

Real engineering version: After entry, C3-00 teaches role shift, Linux/WSL2, shell, Git, venv/dependency pinning, modules/packages/project structure, config/secrets, logging, tests, CLI, Docker and idempotence/runbooks from zero.

7. Follow with me once

Open B06_FINAL_GATE_RULES_RC1.pdf before starting only to understand allowed materials and evidence. Then take Gate A from Assessment/Gate_A. Save the complete attempt. Only after that open B06_GATE_A_REVIEW_PROTECTED_RC1.pdf. If PASS, complete the readiness scorecard. If repair is required, follow the named route. Gate B/C remain sealed unless explicitly assigned.

8. What you should see

The scorecard stays NOT READY until every readiness floor and the Final Gate condition are satisfied. The C3_00_Preview sheet marks all 12 incoming engineering-foundation topics as "No - taught in C3" under pre-entry requirement.

9. Why it worked

Separate floors prevent averaging away critical weaknesses. Protected reassessment prevents learning the exact answer instead of proving the skill. The preview matrix prevents the Bridge from expanding into Course 3.

10. Common beginner mistakes

Marking READY from percentage completion; opening Gate B for extra practice; averaging a 2 in Python against a 4 in SQL; treating C3-00 Linux/Git/Docker as missing Bridge content; erasing failed attempts after a later pass.

11. If stuck

Use the scorecard row that failed. Follow its Repair_Routing entry. Repair only that competency on changed evidence, then take the reassessment route actually assigned by the protected review.

12. Small version alone

Given a mock readiness profile with one weak area, decide READY/NOT READY, name the exact repair module, and explain why no other module should be restarted.

13. Validate

Entry status is READY only when SQL/Python/modeling/warehousing/data quality/independent analysis ≥ 3 ; Cloud/Fabric and professional ownership ≥ 2 ; Final Gate PASS; no unresolved critical blocker; and integrated capstone evidence exists.

14. Teach it back

Explain the difference between "Bridge complete", "C3 entry ready", and "job-ready Data Engineer" in plain language. Then name three C3-00 topics you are explicitly allowed to learn after entry.

15. Protected review + changed retry

Save the Gate attempt first. Review only the attempted variant. Perform targeted repair on changed evidence. Gate B/C are full reassessments, not practice sheets, and open only when assigned. Attempt history remains append-only.

16. Evidence / mastery + next route

Required: protected Gate pass, completed readiness scorecard, repair history if any, and C3 handoff. PASS -> begin C3-00. NOT READY -> exact targeted repair. This course never declares you already job-ready as a Data Engineer.

B06 Quick Reference

Course 3 entry floors

Area	Floor
Advanced SQL	>=3/4
Python automation	>=3/4
Data modeling	>=3/4
Warehousing	>=3/4
Data quality	>=3/4
Independent analysis	>=3/4
Cloud/Fabric	>=2/4
Professional ownership	>=2/4
B06 Final Gate	PASS

What C3-00 teaches next - not pre-entry requirements

C3-00-L01	Role shift: analyst scripts -> engineering systems
C3-00-L02	Linux/WSL2 shell, paths and filesystem
C3-00-L03	Shell pipes, redirection and process basics
C3-00-L04	Git branches, commits and pull-request workflow
C3-00-L05	Python virtual environments and dependency pinning
C3-00-L06	Python modules, packages and project structure
C3-00-L07	Configuration, environment variables and secret boundaries
C3-00-L08	Structured logging and exception handling
C3-00-L09	Unit tests and integration-test boundaries
C3-00-L10	CLI design and reusable data utilities
C3-00-L11	Docker and container mental model
C3-00-L12	Idempotence, reproducibility and runbooks

Automatic blockers

Silent/unreconciled loss or duplication; trusted publish after a critical quality failure; wrong historical answer from incorrect current-state substitution; unsafe rerun; hard access/network/freshness constraint ignored; failure cannot be recovered/explained; protected attempt contaminated by solution copying.

PASS does not mean job-ready DE

PASS means you are ready to begin Course 3. C3 adds engineering environments, ingestion/state, dbt/transformation engineering, orchestration, lakehouse, Spark, streaming, Fabric implementation, DataOps and a production-style capstone.

Next route

C3 ENTRY READY -> begin C3-00. One weak area -> repair exact competency only. Never restart all of Course 2B for one weak area.